

LAND REFORMS, GREEN REVOLUTION AND CROPPING SYSTEMS

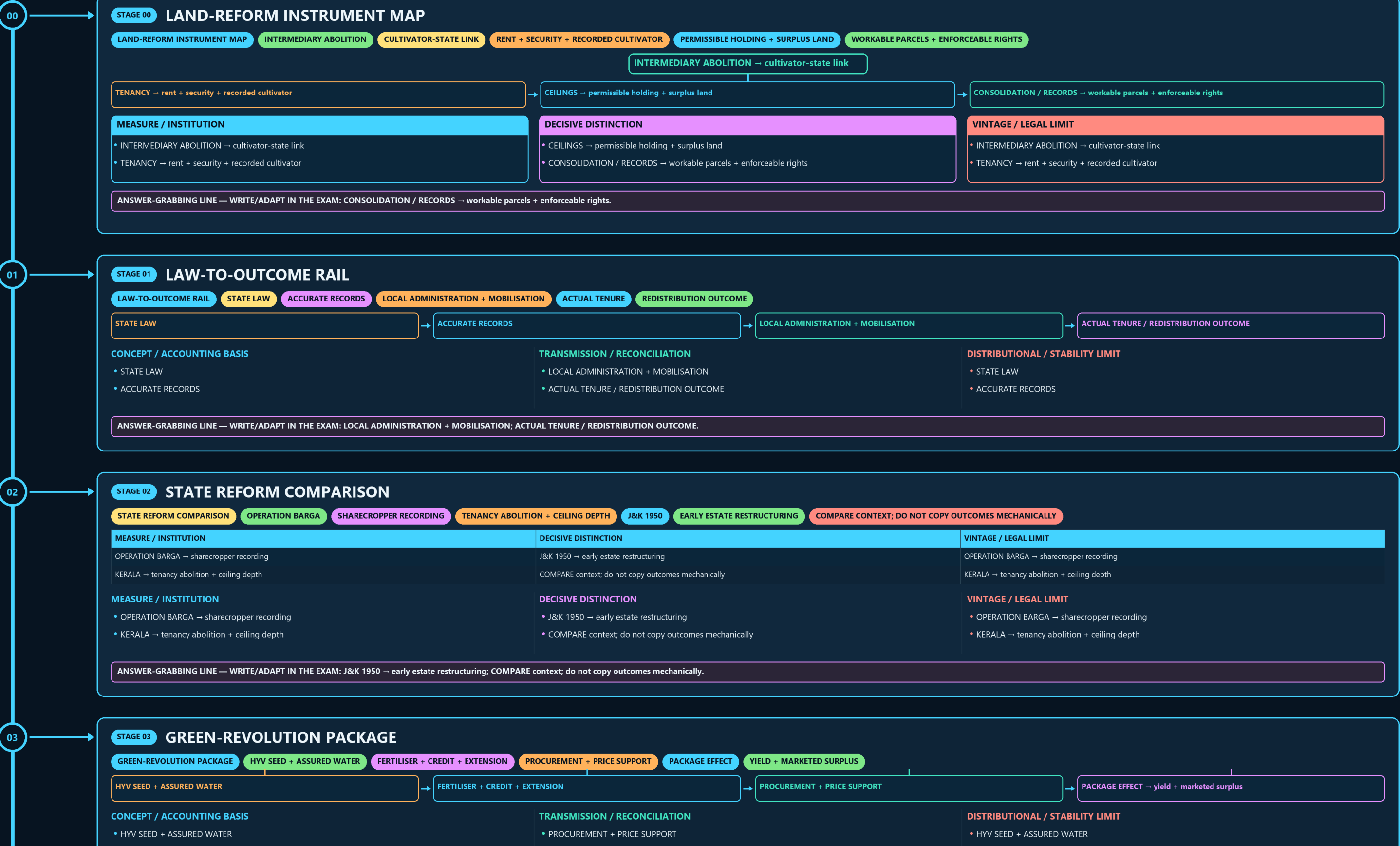
LAND-REFORM INSTRUMENT MAP → LAW-TO-OUTCOME RAIL → STATE REFORM COMPARISON → GREEN-REVOLUTION PACKAGE → CONCENTRATION LEGACY → CROPPING METRICS FORK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



KERALA → tenancy abolition + ceiling depth

MEASURE / INSTITUTION

- OPERATION BARGA → sharecropper recording
- KERALA → tenancy abolition + ceiling depth

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: J&K 1950 → early estate restructuring; COMPARE context; do not copy outcomes mechanically.

COMPARE context; do not copy outcomes mechanically

DECISIVE DISTINCTION

- J&K 1950 → early estate restructuring
- COMPARE context; do not copy outcomes mechanically

KERALA → tenancy abolition + ceiling depth

VINTAGE / LEGAL LIMIT

- OPERATION BARGA → sharecropper recording
- KERALA → tenancy abolition + ceiling depth

03

STAGE 03

GREEN-REVOLUTION PACKAGE

GREEN-REVOLUTION PACKAGEHYV SEED + ASSURED WATERFERTILISER + CREDIT + EXTENSIONPROCUREMENT + PRICE SUPPORTPACKAGE EFFECTYIELD + MARKETING SURPLUS

HYV SEED + ASSURED WATER → FERTILISER + CREDIT + EXTENSION → PROCUREMENT + PRICE SUPPORT → PACKAGE EFFECT → yield + marketed surplus

CONCEPT / ACCOUNTING BASIS

- HYV SEED + ASSURED WATER
- FERTILISER + CREDIT + EXTENSION

TRANSMISSION / RECONCILIATION

- PROCUREMENT + PRICE SUPPORT
- PACKAGE EFFECT → yield + marketed surplus

DISTRIBUTIONAL / STABILITY LIMIT

- HYV SEED + ASSURED WATER
- FERTILISER + CREDIT + EXTENSION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PROCUREMENT + PRICE SUPPORT; PACKAGE EFFECT → yield + marketed surplus.

04

STAGE 04

CONCENTRATION LEGACY

CONCENTRATION LEGACYIRRIGATED NORTHWESTRICE SPECIALISATIONFOODGRAIN SURPLUSGROUNDWATER + SOIL + REGIONAL STRESS

IRRIGATED NORTHWEST → WHEAT / RICE SPECIALISATION → FOODGRAIN SURPLUS → GROUNDWATER + SOIL + REGIONAL STRESS

CONCEPT / ACCOUNTING BASIS

- IRRIGATED NORTHWEST
- WHEAT / RICE SPECIALISATION

TRANSMISSION / RECONCILIATION

- FOODGRAIN SURPLUS
- GROUNDWATER + SOIL + REGIONAL STRESS

DISTRIBUTIONAL / STABILITY LIMIT

- IRRIGATED NORTHWEST
- WHEAT / RICE SPECIALISATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FOODGRAIN SURPLUS; GROUNDWATER + SOIL + REGIONAL STRESS.

05

STAGE 05

CROPPING METRICS FORK

CROPPING METRICS FORKCROP-AREA DISTRIBUTIONGROSS CROPPEDNET SOWN AREALLIED SHIFTEACH NEEDS STATED TIME + GEOGRAPHYPATTERN → crop-area distribution

INTENSITY → gross cropped / net sown area → DIVERSIFICATION → crop / allied shift → EACH needs stated time + geography

MEASURE / INSTITUTION

- PATTERN → crop-area distribution
- INTENSITY → gross cropped / net sown area

DECISIVE DISTINCTION

- DIVERSIFICATION → crop / allied shift
- EACH needs stated time + geography

VINTAGE / LEGAL LIMIT

- PATTERN → crop-area distribution
- INTENSITY → gross cropped / net sown area

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTENSITY → gross cropped / net sown area.

06

STAGE 06

DIVERSIFICATION DECISION BOARD

DIVERSIFICATION DECISION BOARDAGRO-CLIMATE + WATEREXPECTED RETURN + VOLATILITYCREDIT + INSURANCE + EXTENSIONSTORAGE + PROCESSING + BUYER ACCESS

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
AGRO-CLIMATE + WATER	CREDIT + INSURANCE + EXTENSION	AGRO-CLIMATE + WATER
EXPECTED RETURN + VOLATILITY	STORAGE + PROCESSING + BUYER ACCESS	EXPECTED RETURN + VOLATILITY

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
• AGRO-CLIMATE + WATER	• CREDIT + INSURANCE + EXTENSION	• AGRO-CLIMATE + WATER
• EXPECTED RETURN + VOLATILITY	• STORAGE + PROCESSING + BUYER ACCESS	• EXPECTED RETURN + VOLATILITY

- PATTERN → crop-area distribution
- INTENSITY → gross cropped / net sown area

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTENSITY → gross cropped / net sown area.

- AGRO-CLIMATE + WATER
- EXPECTED RETURN + VOLATILITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CREDIT + INSURANCE + EXTENSION; STORAGE + PROCESSING + BUYER ACCESS.

- DRYLAND SUITABILITY
- LOWER WATER EXPOSURE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NUTRITION / RESILIENCE VALUE; NEED processing + procurement + demand.

- PROCUREMENT + CHEAP PUMPING
- WATER-INTENSIVE CROP CHOICE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER-INTENSIVE CROP CHOICE; AQUIFER / SOIL STRESS; DIVERSIFICATION BECOMES HARDER.

- VISVESVARAYA → sluice gates + block irrigation
- SWAMINATHAN → semi-dwarf crop adaptation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENGINEERING → assured water systems; CROP SCIENCE → yield response.

09

STAGE 09 SCIENTIST-ENGINEER DISTINCTION

SCIENTIST-ENGINEER DISTINCTION SLUICE GATES + BLOCK IRRIGATION SEMI-DWARF CROP ADAPTATION ASSURED WATER SYSTEMS CROP SCIENCE YIELD RESPONSE

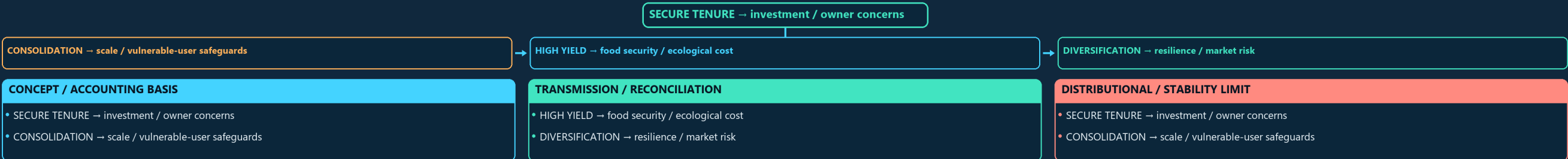
MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
VISVESVARAYA → sluice gates + block irrigation	ENGINEERING → assured water systems	VISVESVARAYA → sluice gates + block irrigation
SWAMINATHAN → semi-dwarf crop adaptation	CROP SCIENCE → yield response	SWAMINATHAN → semi-dwarf crop adaptation
MEASURE / INSTITUTION - VISVESVARAYA → sluice gates + block irrigation - SWAMINATHAN → semi-dwarf crop adaptation	DECISIVE DISTINCTION - ENGINEERING → assured water systems - CROP SCIENCE → yield response	VINTAGE / LEGAL LIMIT - VISVESVARAYA → sluice gates + block irrigation - SWAMINATHAN → semi-dwarf crop adaptation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENGINEERING → assured water systems; CROP SCIENCE → yield response.

10

STAGE 10 AGRARIAN TRADE-OFFS

AGRARIAN TRADE-OFFS SECURE TENURE OWNER CONCERNS VULNERABLE-USER SAFEGUARDS HIGH YIELD FOOD SECURITY ECOLOGICAL COST MARKET RISK

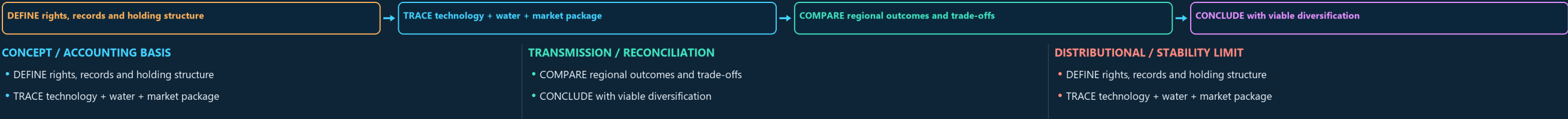


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGH YIELD → food security / ecological cost.

11

STAGE 11 AGRARIAN ANSWER SPINE

AGRARIAN ANSWER SPINE DEFINE RIGHTS, RECORDS AND HOLDING STRUCTURE TRACE TECHNOLOGY + WATER + MARKET PACKAGE COMPARE REGIONAL OUTCOMES AND TRADE-OFFS CONCLUDE WITH VIABLE DIVERSIFICATION



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMPARE regional outcomes and trade-offs; CONCLUDE with viable diversification.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT SECURE TENURE CAN IMPROVE INVESTMENT INCENTIVES, YET POORLY DESIGNED LEASING REFORM CAN SEPARATE OWNERSHIP FROM EFFICIENT CULTIVATION WHILE LAND RECORDS ARE FISCAL, CREDIT, PLANNING AND DISPUTE-RESOLUTION INFRASTRUCTURE GREEN REVOLUTION GAINS CREATED FOOD SECURITY BUT ALSO PATH CROP DIVERSIFICATION REQUIRES DEMAND, PROCESSING, INSURANCE AND LOGISTICS, NOT ALLIED-SECTOR GROWTH CAN STABILISE INCOME BECAUSE PRODUCTION CYCLES AND STATE REVENUE AND LAND DEPARTMENTS

OPTIONAL MODEL / MEASUREMENT DEPTH	VINTAGE, BASKET OR LEGAL LIMIT	COMPARATIVE POLICY NUANCE
- CAUTION: Secure tenure can improve investment incentives, yet poorly designed ceilings may - CAUTION: Leasing reform can separate ownership from efficient cultivation while protecting - CAUTION: Land records are fiscal, credit, planning and dispute-resolution infrastructure, not - CAUTION: Green Revolution gains created food security but also path dependence in water,	- CAUTION: Crop diversification requires demand, processing, insurance and logistics, not price - CAUTION: Allied-sector growth can stabilise income because production cycles and risks differ - State revenue and land departments: implement land records, tenancy and ceiling - Agricultural universities and extension systems: adapt seed and agronomy to local	- Procurement and irrigation agencies: shape crop profitability and regional - Panchayats, co-operatives and farmer organisations: support local records, input - CAUTION: Operation Barga is used to illustrate how political mobilisation and tenancy recording - CAUTION: Punjab-Haryana cereal specialisation shows how irrigation and procurement can lock a

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.